BIOL-1 Practice Test 02 — Answer Sheet

Exam 02 Preparation (Modules 7–11)

Name:		Date:		Score:	

✂ **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice (25 questions) and Part B: Fill in the Blank (10 questions) answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice (25 questions)

1.		10.		19.	
2.		11.		20.	
3.		12.		21.	
4.		13.		22.	
5.		14.		23.	
6.		15.		24.	
7.		16.		25.	
8.		17.			
9.		18.			

Part B: Fill in the Blank (10 questions)

26.		31.	
27.		32.	
28.		33.	
29.		34.	
30.		35.	

BIOL-1 Practice Test 02 — Part C: Short Answer (4 questions)

36. **(Module 7)** Write the complementary DNA sequence AND the transcribed mRNA sequence for the following DNA template strand: **3′-TAC GGG TTA ACG ACT-5′**.

37. **(Module 8 & 9)** Explain the difference between Mitosis and Meiosis in terms of their overall purpose for a multicellular organism like a human. Briefly link how Meiosis connects to Mendel's Law of Segregation.

38. **(Module 10)** Describe how it is possible for a muscle cell and a nerve cell in your body to look and function so differently, despite containing the exact same DNA.

39. **(Module 11)** Choose ONE biotechnology tool discussed in class (e.g., PCR, Gel Electrophoresis, CRISPR, Recombinant Plasmids). Explain what it does and provide one real-world application or example of its use.

Response 1 — Question Number: _____

Response 2 — Question Number: _____

Response 3 — Question Number: _____

Response 4 — Question Number: _____

BIOL-1 Practice Test 02 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 7: Molecular Genetics

1. The enzyme that unwinds the DNA double helix during replication is:
    - A) RNA polymerase
    - B) DNA polymerase
    - C) Helicase
    - D) Ligase

2. Okazaki fragments are synthesized on the:
    - A) Lagging strand
    - B) mRNA transcript
    - C) Leading strand
    - D) tRNA molecule

3. Transcription is the process of converting:
    - A) RNA into a protein
    - B) DNA into RNA
    - C) DNA into a new DNA molecule
    - D) Protein into an observable trait

4. Which part of a primary mRNA transcript is removed during RNA splicing before it leaves the nucleus?
    - A) Introns
    - B) Promoters
    - C) Codons
    - D) Exons

5. A mutation that shifts the entire reading frame of the ribosome is called a:
    - A) Missense mutation
    - B) Silent mutation
    - C) Frameshift mutation
    - D) Point mutation

Module 8: Cellular Genetics (Mitosis & Meiosis)

6. Which phase of the cell cycle is characterized by DNA replication?
    - A) G2 phase
    - B) M phase
    - C) S phase
    - D) G1 phase

7. After a single human cell undergoes meiosis, the resulting cells are:
    - A) 4 haploid, genetically distinct cells
    - B) 2 diploid, genetically identical cells
    - C) 4 diploid, genetically distinct cells
    - D) 2 haploid, genetically identical cells

8. Crossing over, which increases genetic diversity, occurs during:
    - A) Metaphase II of Meiosis
    - B) Prophase of Mitosis
    - C) Prophase I of Meiosis
    - D) Anaphase I of Meiosis

9. Nondisjunction during meiosis can lead to conditions such as Down syndrome, which is caused by trisomy of chromosome:
    - A) 18
    - B) 21
    - C) 23
    - D) Y

10. The p53 protein is heavily involved in:
    - A) Forming the cleavage furrow
    - B) Breaking down the nuclear envelope
    - C) Stopping the cell cycle if DNA is damaged
    - D) Attaching spindle fibers to kinetochores

Module 9: Inheritance Genetics

11. The physical appearance or observable trait of an organism is its:
    - A) Genotype
    - B) Phenotype
    - C) Karyotype
    - D) Allele

12. In a monohybrid cross between two heterozygous parents (Aa × Aa), what is the expected phenotypic ratio for a completely dominant trait?
    - A) 1:2:1
    - B) 3:1
    - C) 9:3:3:1
    - D) 1:1

13. A true-breeding red snapdragon is crossed with a true-breeding white snapdragon, producing offspring that are all pink. This is an example of:
    - A) Incomplete dominance
    - B) Polygenic inheritance
    - C) Sex-linked inheritance
    - D) Codominance

14. Human blood types exhibit which type(s) of inheritance pattern(s)?
    - A) Sex-linked recessive
    - B) Simple dominance only
    - C) Incomplete dominance
    - D) Codominance and multiple alleles

15. Why are sex-linked recessive traits more common in human males than females?
    - A) Males have a stronger Y chromosome.
    - B) The X chromosome in males is larger than in females.
    - C) Males only have one X chromosome, so a single recessive allele determines the phenotype.
    - D) Females lack the genes required for these traits entirely.

Module 10: Epigenetics

16. Epigenetics refers to changes in gene expression that:
    - A) Only occur in prokaryotes.
    - B) Do not change the DNA sequence but can be passed to daughter cells.
    - C) Alter the underlying DNA sequence permanently.
    - D) Are only caused by environmental toxins.

17. DNA methylation generally has what effect on gene expression?
    - A) It increases transcription.
    - B) It causes the DNA to mutate rapidly.
    - C) It has no effect on transcription.
    - D) It decreases or silences transcription.

18. The lac operon in E. coli allows the bacteria to produce enzymes to digest lactose only when lactose is present. This is an example of:
    - A) A sex-linked trait
    - B) A frameshift mutation
    - C) Epigenetic inheritance
    - D) Gene regulation

19. Acetylation of histones usually results in:
    - A) Loosely packed chromatin and increased transcription.
    - B) Immediate degradation of the DNA.
    - C) Tightly packed chromatin and decreased transcription.
    - D) The splicing of introns out of mRNA.

20. The random inactivation of one X chromosome in female mammalian cells results in physical traits like tortoiseshell coloring in cats. The inactivated X chromosome condenses into a structure known as a:
    - A) Centrosome
    - B) Barr body
    - C) Telomere
    - D) Nucleosome

Module 11: Genomics & Biotechnology

21. The primary function of restriction enzymes in biotechnology is to:
    - A) Amplify DNA sequences.
    - B) Analyze protein folding.
    - C) Glue Okazaki fragments together.
    - D) Cut DNA at highly specific sequence sites.

22. Polymerase Chain Reaction (PCR) is used to:
    - A) Rapidly make millions of copies of a specific DNA segment.
    - B) Translate mRNA into a polypeptide chain in a test tube.
    - C) Separate DNA fragments by size.
    - D) Insert foreign DNA into a bacterial plasmid.

23. During gel electrophoresis, DNA fragments move toward the positive electrode because:
    - A) DNA is positively charged.
    - B) DNA has a neutral charge but is pulled by gravity.
    - C) agarose gel attracts DNA chemically.
    - D) the phosphate backbone gives DNA a negative charge.

24. The CRISPR-Cas9 system was originally discovered in:
    - A) Bacteria as a rudimentary cellular immune system against viral bacteriophages.
    - B) Plants to resist fungal infections.
    - C) Human liver cells to repair damaged DNA.
    - D) Viruses as an attack mechanism against hosts.

25. Transgenic organisms, such as bacteria engineered to produce human insulin, are possible because:
    - A) Transcription only works in eukaryotic laboratory cells.
    - B) Bacteria and humans have identical DNA sequences.
    - C) Human genes naturally mutate into bacterial genes.
    - D) All organisms share a nearly universal genetic code.

Part B: Fill in the Blank (10 questions)

Word Bank: Biotechnology, Cytokinesis, Epigenetic, Heterozygous, Lethal, mRNA, Operon, Smaller, Trisomy, Uracil

26. In RNA, the nitrogenous base _____ replaces the Thymine found in DNA.

27. According to the Central Dogma of molecular biology, DNA is transcribed into _____, which is then translated into a protein.

28. The division of the cytoplasm at the very end of mitosis or meiosis is called _____.

29. The presence of three copies of a chromosome instead of the normal two is referred to as a _____.

30. A gene variant that causes death when inherited in homozygous form is known as a _____ allele.

31. An individual with two different alleles for a specific trait (e.g., Bb) is called _____.

32. In gene regulation, a(n) _____ is a cluster of genes under the control of a single promoter, commonly found in prokaryotes.

33. Identical twins have the exact same DNA sequence, but as they age, differences in their gene expression accumulate due to _____ changes.

34. In gel electrophoresis, _____ fragments of DNA travel through the gel matrix faster and farther than larger fragments.

35. The use of an organism, or a component of an organism, to make a product or process for human use is known as _____.
